

A New Family of Operators on Cellular Automata

Joseph Corneli

Abstract

We describe a cellular automaton whose cells do not hold states but *rules*, and an operator that rewrites those rules by acting on the rule's own truth table: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$ for σ a permutation of the eight neighbourhoods. The rule's truth table is itself an eight-cell automaton whose topology is σ — an automaton inside the automaton — giving a family of $8! = 40,320$ operators that is finite, exhaustively enumerable, and exactly classified.

The classification is by cycle parity. A fixed point of P_σ exists if and only if every cycle of σ is even; the number of such σ is $((2n-1)!!)^2$, which for S_8 gives $105^2 = 11,025$ of 40,320, or 27.34%. In a census of the 20,256 mirror orbits, run under a spatial rule-field dynamics, the all-even class is observed to go extinct several times more often than the rest.

The main result concerns where those fixed points *are*. Without exception, across all 40,320 operators, every fixed point that exists has exactly four one-bits: a *balanced* truth table, equivalently Langton's activity $\lambda = 1/2$. This is forced by parity—an alternating colouring assigns each even cycle of length $2m$ exactly m ones, so the total is $8/2 = 4$ necessarily. A balance long associated *empirically* with Langton's edge of chaos is here an exact and elementary consequence of the fixed-point condition. Operators with all-even cycle structure possess these fixed points; operators with any odd cycle possess none. Possessing a fixed point at $\lambda = 1/2$ is not, however, the same as sustaining complex behaviour: P_σ is a bijection, so a fixed point is not by itself an attractor. In the spatial dynamics we run, the all-even class is observed to settle onto its fixed points more readily—and because every fixed point shares the one value, λ cannot distinguish an operator whose field stays diverse from one whose field collapses. For that we use a simple population observable, the count of distinct rules present over time, which in the runs we report separates the rotations that sustain a diverse field from those that do not.

A New Family of Operators on Cellular Automata

The Object

A cellular automaton normally evolves a field of *states* under one fixed rule. Invert that arrangement: let each cell hold a *rule*, and let an operator evolve the field of rules. An elementary cellular automaton (ECA) rule is an eight-bit truth table $\text{bit}[k]$, with k ranging over the eight three-cell neighbourhoods. We study the operator P_σ , parameterised by a permutation $\sigma \in S_8$ that acts on the rule's own table. Writing a rule as a map $g: \{0, \dots, 7\} \rightarrow \{0, 1\}$, the operator sends g to $P_\sigma g$ defined by

$$(P_\sigma g)(j) = \neg g(\sigma^{-1}(j)), \quad \text{equivalently} \quad \text{bit}[\sigma(k)] := \neg \text{bit}[k]. \quad (1)$$

Each output bit is one input bit, at the position σ moves it to, negated; P_σ is thus a permutation of the eight coordinates composed with a global complement. The truth table is thereby itself an eight-cell automaton whose wiring is σ and whose update is Boolean negation — an automaton inside the automaton. Equation (1) defines a family of $8! = 40,320$ operators that is finite, exhaustively enumerable, and, as we show below, exactly classified. Box 1 works through one member of the family bit by bit.

Worked example 1: A propagator by hand

Write an ECA rule in Wolfram's standard neighbourhood order:

neighbourhood	111	110	101	100	011	010	001	000
Rule 110	0	1	1	0	1	1	1	0

Thus Rule 110 is the byte 01101110. Let $\sigma(k) = k+2 \pmod{8}$: each source bit is complemented and written two positions further along. Carrying out the eight writes gives

$$P_{+2}(01101110) = 01100100,$$

which is Rule 100. This changes what the cell does to its black-and-white phenotype. For example, neighbourhood 011 maps to 1 under Rule 110 but to 0 under Rule 100. The propagator has not directly flipped a phenotype cell; it has changed the truth table that the cell will use.

The bare operator acts on one eight-bit rule. In the spatial experiments it is placed inside a two-layer cellular automaton: a field of rules (the *genotype*) governs a field of black-and-white states (the *phenotype*), while neighbouring rules also interact. Box 2 spells out one complete local update; this is the computation behind the space–time diagrams in Figure 1.

Worked example 2: One site, one complete MetaCA step

At time t , site x holds an eight-bit rule $g_x(t)$ and a phenotype bit $X_x(t)$. One ordinary propagator step has three parts.

1. **Express the current rule.** The phenotype neighbourhood is evaluated by the rule at its centre:

$$X_x(t+1) = g_x(t)(X_{x-1}(t), X_x(t), X_{x+1}(t)).$$

2. **Blend neighbouring rules.** For each truth-table position j , inspect the three genotype bits $(g_{x-1}(j), g_x(j), g_{x+1}(j))$. If the two outer bits agree, copy their value. Thus $(0, 1, 0)$ blends to 0. If they disagree, the centre rule decides by treating the triple as one of its ECA neighbourhoods. The eight coordinate-wise decisions form an intermediate rule

$$h_x = B(g_{x-1}(t), g_x(t), g_{x+1}(t)).$$

3. **Propagate the blended rule.** Apply the selected wiring to that intermediate byte:

$$g_x(t+1) = P_\sigma(h_x).$$

The ordinary runs apply P_σ once. The historical Figure 8 accident, replayed in Figure 2, applied its off-by-one propagator twice.

The substrate—a cellular automaton whose cells hold rules rather than states—is not itself new (Corneli & Maclean, 2015); what we study is the *operator* of Equation (1), understood as an enumerated, exactly classified family. That 2015 paper found a single member of this family, encountered through a programming error and not recognised as one

of many; we now give the family an essentially complete characterisation. The canonical symmetry group of ECA rule space has order four (reflection $\times$ complement, collapsing 256 rules to 88 classes) (Schaller & Svozil, 2025); it is not S_8 , and the negation-carrying, σ -parameterised family of Equation (1) does not appear in it. Rule-space permutation as a parameterised, enumerated family with a derived fixed-point structure does not appear in the literature we have surveyed; the nearest neighbours augment cellular automata with state-dependent feedback (Pavlic et al., 2014) rather than permuting the rule itself. Two established genres are adjacent but distinct: per-cell rules (non-uniform or ν -CA (Dennunzio et al., 2011)) and rules that change during evolution (rule-changing or programmable CA (Mori et al., 1998)). A referee should note the trap in the first: a ν -CA drawing rules from a finite set is equivalent to a uniform CA on the product alphabet (Dennunzio et al., 2011), so the substrate is not where novelty lives. It lives in the fixed-point structure of the operator, which is what the rest of this paper is about.

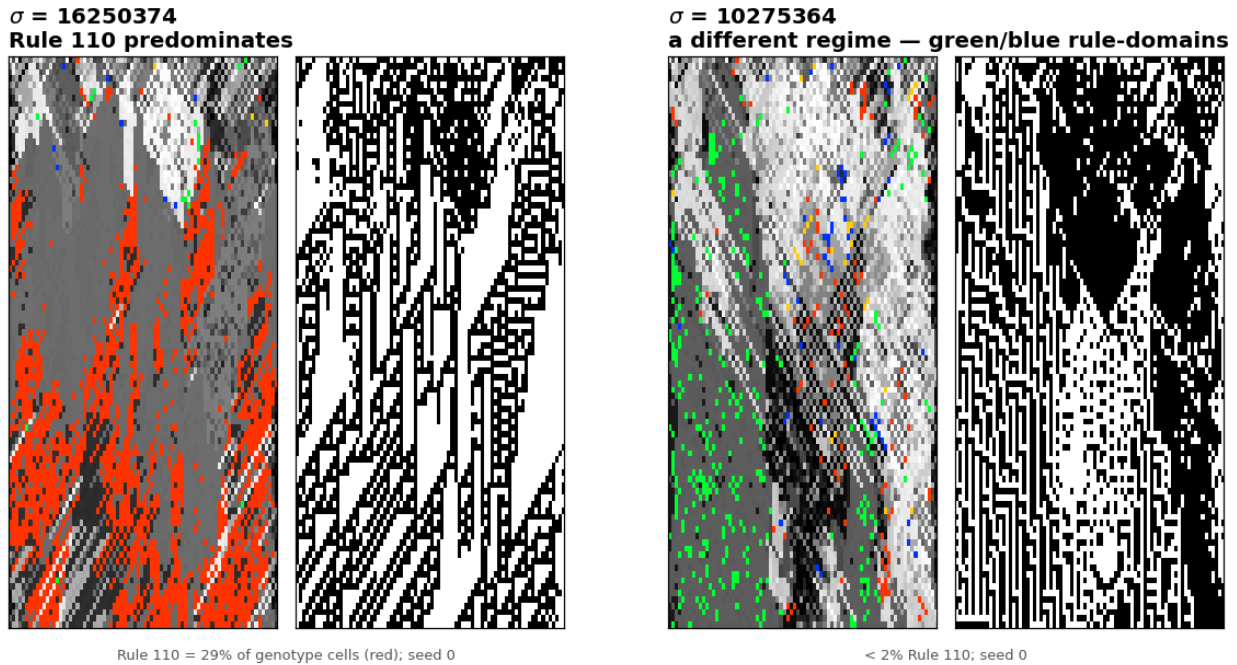
The Classification

Fixed points exist iff every cycle is even

Theorem 1. *P_σ has a fixed point (a fixed rule) if and only if every cycle of σ has even length. Since a one-cycle has odd length, this already requires σ to be fixed-point-free.*

A fixed rule is a g invariant under (1), i.e. a σ -equivariant Boolean colouring with $g(\sigma k) = \neg g(k)$. Follow a cycle of length L : equivariance forces $g(k) = (\neg)^L g(k)$, and $(\neg)^L$ is the identity exactly when L is even; conversely, colour each orbit by the parity of its distance from a chosen representative. A one-cycle of σ ($L = 1$) is odd and so admits no such colouring, which is why even cycles are required throughout.

Box 3 shows the theorem on the offset-+2 operator. The same small calculation also makes the later 2^k count and forced $\lambda = 1/2$ balance visible before we state them generally.

**Figure 1**

Two operators from the family, each shown over time as genotype (256-colour, left) and phenotype (black and white, right). Under $\sigma = 16250374$ the genotype field comes to be dominated by ECA Rule 110—the red cells, 29% of the genotype at the representative seed (35% over six seeds)—and the phenotype accordingly displays Rule 110's characteristic gliders; under $\sigma = 10275364$ a different regime of rule-domains forms (green and blue, under 2% Rule 110). The operator does not merely stir rule space: it drives it toward particular, computationally distinguished rules. Genotype colours are the legacy engine's own.

Worked example 3: Why even cycles work

For offset +2 on eight truth-table positions,

$$\sigma = (0\ 2\ 4\ 6)(1\ 3\ 5\ 7).$$

A fixed rule must change colour at every arrow because $g(\sigma k) = \neg g(k)$. Around either four-cycle it can therefore read 0101 or 1010: after four negations it returns consistently to its starting bit. The two cycles can be chosen independently, giving

$$2^2 = 4 \quad \text{fixed rules: } 51, 102, 153, 204.$$

Each contains four zero-bits and four one-bits. An odd cycle fails for the same elementary reason: after an odd number of negations it returns to the starting position demanding the opposite of the bit with which it began.

This is the signed-cycle fixed-point theorem of Boolean-network theory (Aracena, 2008) specialised to (1): our operator is a Boolean network whose signed interaction digraph is σ with every arc negative, a negation-cycle of length ℓ has sign $(-1)^\ell$, and an isolated negative cycle admits no fixed point while a positive (even) one admits exactly two. We contribute a machine-checked proof of the specialised statement in Lean 4 / Mathlib.¹

The closed form

The number of $\sigma \in S_{2n}$ with all cycles even is $((2n-1)!!)^2$, with exponential generating function $\sum_n ((2n-1)!!)^2 x^{2n} / (2n)! = 1/\sqrt{1-x^2}$ (the “set of even cycles” construction). For S_8 this is $105^2 = 11,025$ of 40,320, or 27.34%. The count is classical: it is OEIS Foundation Inc. (2011) and appears as Stanley (1999, Problem 5.34(c)) and in Riordan (1968). We claim it only as the enumeration of our operator family, not as a new identity.

¹ `DarkTower/Patterns/Propagator.lean`, theorem `hasAlternatingColouring_iff_cycleType_even`;
`lake build clean`, no `sorry` or `axiom`.

Fixed bytes number 2^k

An all-even σ with k cycles has exactly 2^k fixed bytes; any σ with an odd cycle has none. Each even cycle admits exactly two alternating colourings, chosen independently across cycles, whence 2^k ; the zero case is Theorem 1. This too is machine-checked.² Summed against these weights the family is consistent: $\sum 2^k$ over all-even σ equals $(2n)! = 8! = 40,320$, the total number of fixed bytes across the family.

Every Fixed Point Is Balanced

Langton's λ measures a rule's *activity*: the fraction of its neighbourhood entries that map to the non-quiescent state (Langton, 1990). For an eight-bit ECA table it is the Hamming weight over eight, $\lambda = \|r\|_1/8$, running from 0 (every neighbourhood dies) through $1/2$ to 1. Langton's thesis was that complex, computation-supporting behaviour concentrates near a critical value of λ . That transition was demonstrated most clearly for larger rule tables (notably $K = 4$, $N = 5$); for the two-symbol, three-neighbour case of the ECAs, Langton is explicit that λ is only roughly correlated with dynamical behaviour, and later work is cautious about the correlation (Mitchell et al., 1993). Our theorem below is about the value $\lambda = 1/2$ itself—*balance*, a property of the truth table—and we keep that exact statement separate from the empirical edge-of-chaos programme it evokes.

Every fixed point has $\lambda = 1/2$

Theorem 2. *Every fixed point of every operator in the family has exactly four one-bits; equivalently, Langton's activity parameter $\lambda = 4/8 = 1/2$.*

By Theorem 1 each cycle has even length $2m$; an alternating colouring assigns such a cycle exactly m ones; summing over cycles gives $(\sum L)/2 = 8/2 = 4$. Exhaustive enumeration confirms it: across all 40,320 operators and all their fixed points, every one has Hamming weight four, with zero exceptions. The statement is machine-checked for Fin 8.³

² `alternatingColouring_card_eq_two_pow_cycleType_card`.

³ `alternatingColouring_true_card_eq_four`: the true-cells and false-cells are put in bijection by σ ,

We are deliberate about the standing of this result. Langton’s critical value was obtained empirically, as a statistic over rule space (Langton, 1990), and its status as an order parameter was contested: Mitchell et al. (1993) refuted Packard (1988)’s specific genetic-algorithm clustering claim, not the definition of λ , which survived as one axis among others such as Wuensche’s Z (Wuensche, 1999). There is a folklore intuition for why $1/2$ is special — binomial sampling variance $\lambda(1 - \lambda)$ is maximal there — and it is not ours; and it has been noted that λ alone does not fix a rule’s dynamical class, distinct classes coexisting at a single value (Sakai & Kanno, 2002). What Theorem 2 adds is narrower and firmer than any of these: under the complementation in (1) the value $1/2$ is *forced* as an algebraic matter, because any alternation-under-complement colouring is balanced by parity. We claim balance, exactly; the connection of balance to edge-of-chaos dynamics we treat as interpretation, not as part of the theorem.

The cast: which rules survive, and how many

Under an all-even σ , the fixed points of P_σ —the rules it maps to themselves—number 2^k , where k is the number of cycles of σ (*The Classification*): each even cycle admits two alternating colourings, chosen independently across cycles. In the spatial runs these are the rules onto which the field is observed to settle. Every one has $\lambda = 1/2$ (Theorem 2), so the observed terminal population is a cast of *distinct* rules—distinct truth tables—sharing a *single* activity.

The smallest nontrivial case of the count is a single cycle, $k = 1$: two survivors. The 2015 paper’s Figure 8 already exhibited a two-rule residual (Figure 2): a random field of tens of distinct rules dies off to an alternating population of just two, rules 42 and 170. Fittingly, the programming error that produced the family is what pins its residual at two—the erroneous mutation froze a bit, splitting the field into the cells that can reach 42 and those that can reach 170—so the phenomenon was on the page before the 2^k count explained it.

forcing each to number four.

That die-off curve—the count of distinct rules present over time—is a natural population-level companion to λ . Where λ is a scalar assigned to a single rule, this curve is a property of the whole field’s trajectory; and where λ is uninformative here—every fixed point sits at $1/2$ —the curve still varies, because it measures the population rather than the point. We use it cautiously, as an observable rather than an established order parameter, to tell the rotations whose fields stay diverse from those whose fields collapse (*Fixed Points Are Not Attractors*, Figure 6).

Diversity of the observed population and singularity of its activity thus coexist; the metric measures we tried do not separate the two (*Why Distance Geometry Cannot See This*).

A dead genotype can carry a live phenotype

The balanced shell is a definite object: exactly $\binom{8}{4} = 70$ rules carry four active outputs, and by Theorem 2 the absorbing set of every operator in the family lies wholly inside it. Collapse is therefore not merely a loss of diversity but a fall onto this particular 70-rule shell—the field dies *onto the edge*, at $\lambda = 1/2$, always.

Balance is necessary for a genotype to freeze and silent about what the frozen rule then does. Run each of the 70 as an ordinary elementary CA and they split: 32 are phenotypically live—sustained change under iteration, the additive rules 90, 105, 150 and the class-IV rule 54 among them—while the rest fall quiescent. The two halves share a single λ ; this is, inside our family and by construction rather than by survey, the coexistence of dynamical classes at one activity value that has long been urged against λ as an order parameter (Sakai & Kanno, 2002). A field can thus die in its genotype while the surviving rule keeps its phenotype in motion.

Figure 3 is such a field, built to specification. To seat a collapse on a chosen live rule R it suffices to pair R ’s four active positions with its four inactive ones as an involution: four transpositions, an all-even operator that must collapse (Theorem 1) and that holds R among its fixed points. Carried out for Rule 54 this yields $\sigma = [2\ 3\ 0\ 1\ 5\ 4\ 7\ 6]$;

run from a random field it drives the genotype to near-uniformity—two rules survive—while the phenotype never settles. The field has died onto Rule 105, another live tenant of the same shell, whose additive dynamics keep the black-and-white panel full of structure. Genotypic death and phenotypic death are not the same event.

One caveat marks the open edge of this. An involution fixes not its target alone but the whole shell compatible with its pairing, and which tenant a random field actually reaches is set by the basins, not by the construction: we aimed at 54 and the dynamics preferred 105. Existence is settled—live survivors are constructible—but *steering* a field onto a named rule is not, and is the natural next question.

The Census

Parity predicts death across all 20,256 orbits

Theorems 1–2 are statements about fixed points; we also ran the family as a spatial dynamics to ask whether the parity dichotomy is reflected in observed behaviour. Reducing the family by the mirror symmetry gives 20,256 orbits. In our runs—a field of rules on a one-dimensional lattice, evolved from random initial rules and scored for extinction of phenotype activity—the all-even class goes extinct several times more often than the rest, with the tendency increasing in the number of cycles k . We report this as an observed trend, not a measured law: the present runs fix a single lattice size, boundary condition, initial distribution, and horizon, and we have not yet tabulated deaths and denominators by class and by k with seed-level uncertainty—which a quantitative claim would require. Note also that classifying the enumerated permutations by cycle parity merely recovers the combinatorial 27.34%; the dynamical content is the *death-rate difference* between the classes, not the class sizes. Figure 4 illustrates the dichotomy with five representative operators rather than summarising all 20,256.

A prospective test: reduced rotate+2

To guard against reading structure into a completed enumeration, we fixed a prediction before running the case. The reduced rotate+2 operators act as a single even

cycle on a subset of positions and leave the remaining positions as one-cycles; by that leftover they are *not* all-even, so by Theorem 1 they possess no fixed point and should fall on the no-fixed-points side of the dichotomy—no rule for the field to settle onto, and correspondingly less extinction. They do (Figure 4, right): they stay diverse rather than dying off. Absence of fixed points forbids settling onto a single rule; it does not forbid short-period behaviour, and we claim only the observed persistence of diversity, not literal aperiodicity.

Fixed Points Are Not Attractors

Possessing a fixed point (Theorem 1) is not the same as reaching one, and P_σ does not by itself move a rule toward $\lambda = 1/2$. Because P_σ complements every bit, $\lambda(P_\sigma g) = 1 - \lambda(g)$: the operator *reflects* activity across $1/2$, exactly preserving the distance $|\lambda - 1/2|$ rather than contracting toward it. And because P_σ is a bijection on bytes, a generic rule lies on a periodic orbit and simply cycles—its fixed points are fixed, but they are not attractors and have no basins. The only rules pinned at $\lambda = 1/2$ are the fixed points themselves. Whatever pull toward balance the *spatial* field exhibits must therefore come from the coupling between cells and from the interrupter below, not from the bare map.

The rotation family makes this concrete. Every non-trivial wrap rotation is all-even and so possesses fixed points (Theorem 1), yet in the spatial dynamics only offset ± 2 — $\gcd(2, 8) = 2$, two 4-cycles—sustains a visually complex, high-diversity field (Figure 5), persisting to roughly a hundred generations; the remaining rotations ($\gcd 1$ and $\gcd 4$) collapse to near-uniform fields within a few dozen. Possessing a fixed point and having the field stay near it are different properties, and only the second is visible in a drawing.

This last distinction can be *measured*, with the distinct-rule population curve as a simple instrument. Every rotation begins at peak diversity—some seventy distinct rules in a random field—but the distinct-rule count then diverges sharply by cycle structure (Figure 6): offset ± 2 hold around thirty distinct rules through the longest runs we made

(400 generations), while every gcd 1 and gcd 4 rotation collapses to one or two within sixty. The persistent diversity of the gcd 2 operators and the quick collapse of the rest—visible by eye in Figure 5—are thus separated by more than an order of magnitude in this one scalar. We do not claim the count is an order parameter in the statistical-mechanics sense; it is a useful summary of what the drawings show, on the two seeds per operator that we ran.

Sustained structure requires an *interrupter*—a step that occasionally declines to propagate. Composed with such a step (a neighbour-blend that copies where neighbours agree, or a hold-when-active gate), the coupled dynamics no longer settles: instead of a single frozen byte or a runaway cycle, the field holds a diverse population of rules, many of them at $\lambda = 1/2$. The coordinate-wise blend in Box 2 is the interrupter used in the ordinary spatial runs. This is what resolves the apparent tension between the census and Figure 6: they are the same operator under two *couplings*. In the census runs, left to settle, the field under rotate+2 collapses onto a few of its fixed points; with the interrupter it is held off them and keeps tens of rules. Settling and sustained diversity are two behaviours of one operator, selected not by the bare algebra but by the coupling.

Propagators do not compose usefully with one another, and this negative result shapes what composition can do here. Each operator of Equation (1) negates every bit— $P_\sigma(g)[j] = \neg g[\sigma^{-1}(j)]$, a permutation of the eight coordinates followed by a global flip—so under composition the flips cancel in pairs. The product $P_\sigma \circ P_\tau$ is a *pure* coordinate permutation, with no negation left in it—itsself a bijection, hence (like every map in the family) not an absorber: it fixes 2^c bytes for c cycles but sends a generic byte around a cycle, and it carries none of the forced $\lambda = 1/2$ structure that made the single operator interesting. An even number of propagators composes to such a bare permutation, an odd number back to a single propagator $P_{\sigma\tau\dots}$; either way the composite contains nothing the individual operators did not. The family is flat under its own composition. We searched composites of two and three operators for emergent structure and found none—the negation that gives one operator its balanced ($\lambda = 1/2$) fixed point is exactly

what a second operator cancels—so the compositions that *do* yield something new must pair a propagator with a step of a different kind.

One such composition is worth exhibiting. Taking the offset-+2 propagator—the gcd 2 operator of Figure 5—and composing it with a construction step that reads the phenotype, matches it against four local template candidates, and falls back to the all-zero rule when none matches produces a *river*: a coherent band of alternating phenotype that drifts through a disordered background and reappears in all six seeds we tried (Figure 7). Where the bare rotation collapses or cycles, this composition holds an off-critical structure for the length of every run—its trajectory varying with the seed, its presence not. We report these two mechanisms here; a fuller catalogue of composed operators is future work.

Why Distance Geometry Cannot See This

The classification is a conjugacy invariant of the operator σ —its cycle type—while the complexity and compression measures we tried are computed from a single rule’s bits or from a single space–time trajectory. A quantity read off one field’s history does not have access to σ ’s cycle type, so we should not expect it to recover the parity classification, and in our tests it did not. The smallest surviving cast is small but not degenerate—an all-even σ with a single 8-cycle ($k = 1$) has $2^1 = 2$ fixed points, since a permutation of eight elements has at least one cycle—yet a population-diversity count reads such a collapsed field as nearly dead though it is maximally selected; a frozen periodic genotype scores a lower statistical complexity C_μ (Crutchfield & Young, 1989) than Rule 110; gzip codelength rates a self-similar rule as noise. These observables are not all metrics in the same sense, and we do not claim a general impossibility theorem—only that the standard per-trajectory measures we tried do not separate the parity classes, and that this is unsurprising because the classifying property lives on the operator, not on the states. The classification is a theorem *because* it is not a per-trajectory statistic.

Discussion

The survey of Figure 5 exposes a sharp behavioural split within a family of provably fixed-point-bearing operators: only offset ± 2 sustains a high-diversity, visually complex phase, while the rest collapse to near-uniform fields. That surviving phase carries the visual signature long associated with the *edge of chaos* (Langton, 1990), and a reader is entitled to ask whether the resemblance is more than skin deep. It is, but only in a specific and measurable sense, and naming that sense is the purpose of this section.

Two notions travel under the one phrase. The first is *global*: a system tuned by a control parameter to a critical point between an ordered and a disordered phase—Langton’s λ , which *Every Fixed Point Is Balanced* shows is silent here, every fixed point sitting at $\lambda = 1/2$ by Theorem 2. The second is *local*: within a single realisation, ordered and disordered regions coexist in space, and the edge of chaos is the *interface* between them—the domain walls of computational mechanics (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001), where structure propagates and interacts. It is the second that this object supports.

To expose the local structure we colour the genotype field not by rule identity but by each rule’s *intrinsic dynamical character*: a scalar chaos score obtained by running that rule alone as an elementary CA and measuring its *dynamical* activity—how much its field keeps changing over time, a run-time quantity distinct from the static truth-table balance λ of *Every Fixed Point Is Balanced*. Ordered rules score near 0; complex rules such as Rule 110 near 0.4; chaotic rules 0.5 and above. Under this tint (Figure 8) the field resolves into coexisting domains of distinct dynamical character—ordered, complex, and chaotic—separated by definite boundaries rather than a uniform turbulence. Which regimes coexist depends on the operator: offset +2 pairs ordered with chaotic, whereas $\sigma = 16250374$ pairs ordered with the complex Rule 110 that comes to dominate its field (*Conclusion*). Thinning the boundary between ordered and active regions to a filament skeleton recovers the “separating curve” directly from the data.

The interface is not a clean line but a *fractal*. Box-counting yields a scale-free dimension (clean power laws, $R^2 > 0.99$) of $D \approx 1.5$ for offset +2 and $D \approx 1.7$ for the phenotype-reading river of Figure 7. Two features lift this above curiosity (Figure 9). First, across system sizes $L = 128$ to 768 the dimension *stabilises* rather than drifting, and the coexistence fraction is *size-independent* (≈ 0.4 at every L): the operator neither orders-out nor chaos-out as it grows, but holds a scale-invariant mixture. Second, the behaviour cleanly separates sustaining from collapsing operators: offset +4, superficially similar in isolation, loses its chaotic phase *entirely* at $L \geq 256$ —the interface does not simplify, it disappears. The edge of chaos, locally understood, is thus a persistent, scale-invariant geometric feature of the sustaining operators and demonstrably absent from the collapsing ones.

We deliberately stop short of two stronger claims. We do not assert a global critical transition; consistent with *Limits*, the honest reading remains a stable coexistence rather than a demonstrated critical point, and the interface analysis here is exploratory—a handful of seeds, on the fixed-boundary (line) variant rather than the wrapped lattice of Figure 5; a boundary-matched, more heavily seeded finite-size collapse is the natural next step. Nor do we claim computation: direct injection–decode probes recover information storage and short-range transmission on these fields, but no nonlinear modification under the conditions tested. The edge of chaos we defend is the structural one—a scale-invariant order/chaos interface—not the computational one.

The local picture also ties back to what makes these operators new. The genotype–phenotype coupling is not incidental to the interface: the phenotype-reading river sustains a *denser* fractal boundary ($D \approx 1.7$) than the feedforward offset +2 ($D \approx 1.5$)—a difference visible at the interface, though whether the coupling *drives* it we do not settle here. And it points at a sequel. If the filaments are domain walls, they are candidate *wires*: the natural computational question is not which rules compose them—in our data the complex rules are not enriched on the interface—but whether information is *transported*

along them and *combined where they collide* (Mitchell et al., 1993). Whether computation flows down the edge-of-chaos lines of these diagrams is, we think, a paper of its own.

Limits

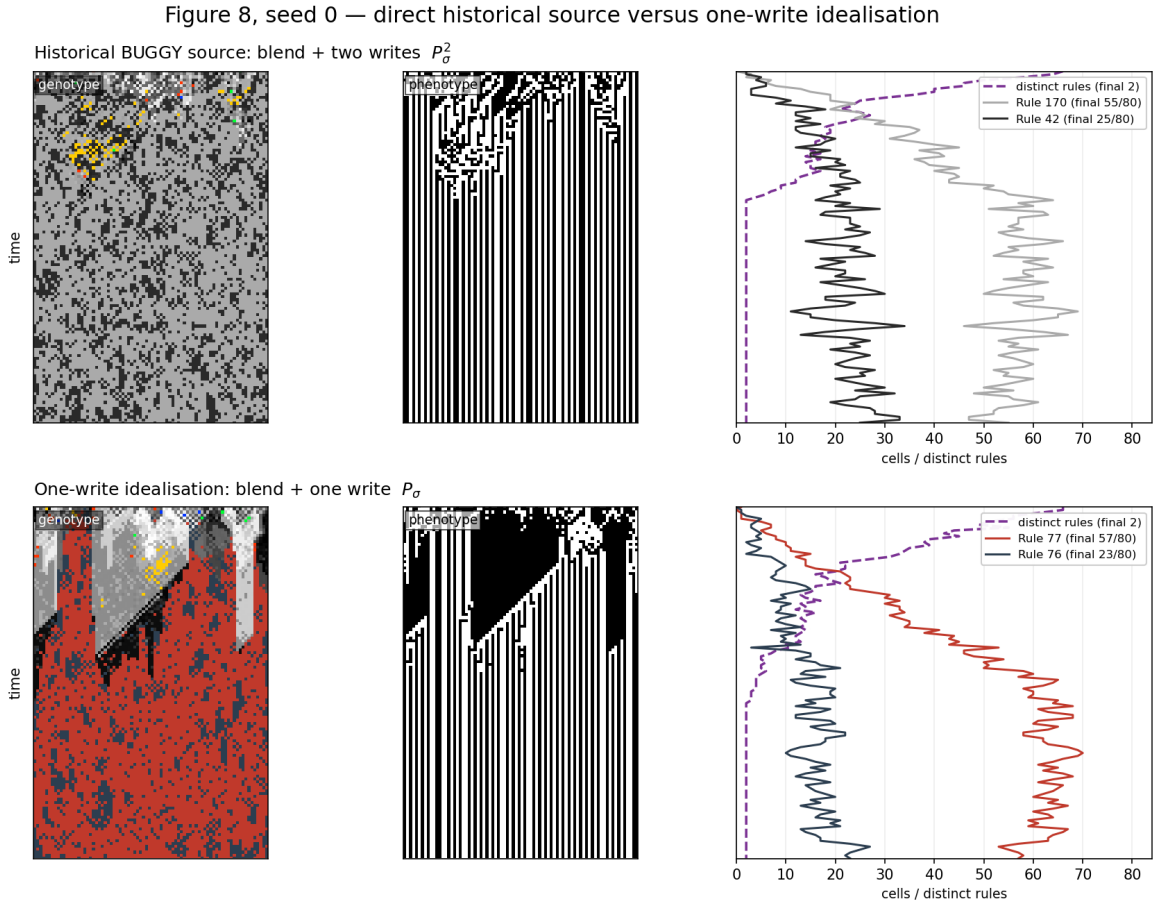
The theorems are the Boolean case and are exact; the dynamical claims are narrower and empirical. The census and the sustained-diversity behaviour are reported for a single spatial coupling (S_8 , radius one, one value map) at limited seed counts, and invite a fuller treatment; we use “edge of chaos” descriptively, not as a measured critical point.

Discussion takes up the local, structural reading and reports an exploratory interface-dimension statistic under its own stated caveats (few seeds, fixed-boundary variant); beyond it we report no perturbation-spreading or entropy-rate statistics. Because the dynamics is stochastic, every such figure is given both as a fixed *representative* seed and, where a scalar is quoted, as a fixed *six-seed sample* (mean and range); both are reproducible from the pinned seeds through the accompanying package. The substrate the object sits in is not itself new (*The Object*); λ is one heuristic among several (Wuensche, 1999), and we make no claim that balance implies edge-of-chaos dynamics—only that here balance is *derived*. What is new, and what we rest the paper on, is the object of Equation (1) and the fixed-point structure that forces every fixed rule to be balanced, $\lambda = 1/2$, as a matter of proof.

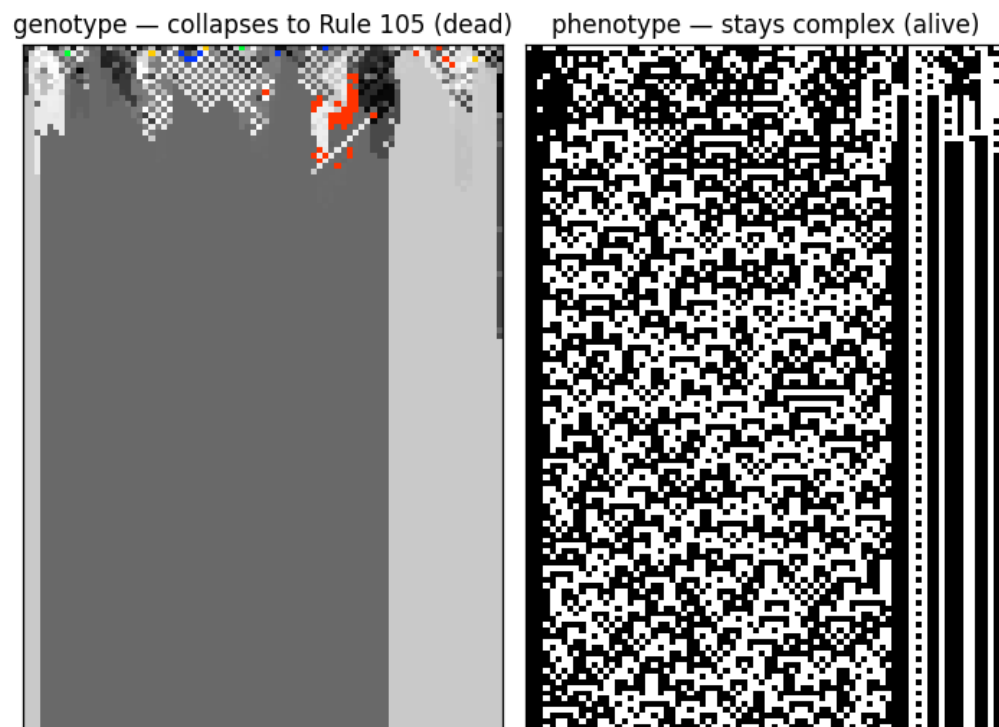
Conclusion

The characterisation promised at the outset (*The Object*) has two halves. The first is static and exact: which operators possess fixed points (the all-even ones), how many (2^k), and where they lie (balanced, $\lambda = 1/2$, always). The second is empirical and tentative. Because every fixed point sits at $\lambda = 1/2$, that parameter is silent on the difference between an operator whose field stays diverse and one whose field collapses; the distinct-rule count is not, and in our runs it separates the two across the rotation family by more than an order of magnitude (Figure 6). The same observable offers a suggestive reading of the search that produced the family, reported both as a representative run and

as a fixed six-seed sample (*Limits*). The operator of Figure 1 that comes to be dominated by Rule 110—a universal rule, the naive prize of a search for computational intelligence—concentrates the field onto it and pays for it in diversity. At the representative seed, 29% of its genotype cells are Rule 110 and it sustains 17 distinct rules, against a foil that holds Rule 110 below 2% and sustains 20; over the six-seed sample the ordering is unchanged—Rule 110 35% (range 27–40%) versus 2% (1–2%), sustained diversity 15 versus 24. On this pair, recovering a universal rule and sustaining a diverse population pull against each other—a suggestion worth testing at scale, not a law we have established.

**Figure 2**

The family's accidental first sighting, separated from its later idealisation (seed 0, width 80). Top: a direct replay of the historical Figure 8 source. Its neighbour blend is followed by two independent calls to the erroneous mutation, equivalently two writes by $\sigma = [0, 0, 1, 2, 3, 4, 5, 6]$. In both rows the distinct-rule count is purple and dashed. Here it falls from tens to two; rules 170 (light grey) and 42 (dark grey) partition the field, with population traces matching their genotype colours. Bottom: the later one-write propagator idealisation, which applies the same σ only once after blending. Its terminal genotypes, rules 77 (red) and 76 (dark), are colour-coded to distinguish their adjacent greyscale rule numbers. This lower panel isolates the propagator cleanly, but is not a one-for-one reproduction of the 2015 figure. Genotype, phenotype, and population share the time axis within each row.

**Figure 3**

A genotypically dead, phenotypically live field. The all-even operator $\sigma = [2\ 3\ 0\ 1\ 5\ 4\ 7\ 6]$, built to fix the balanced shell containing Rule 105, collapses the genotype (left, 256-colour) to near-uniformity within a few dozen rows—two rules survive—yet the phenotype (right) never settles: the field has died onto Rule 105, an additive $\lambda = 1/2$ rule whose dynamics keep it complex. Seed 1, $W = 80$, $T = 120$.

CYCLE PARITY DICHOTOMY: a Boolean fixed point exists iff every cycle is even (Lean: `bool_fixed_exists_iff_hasAlternatingColouring`)

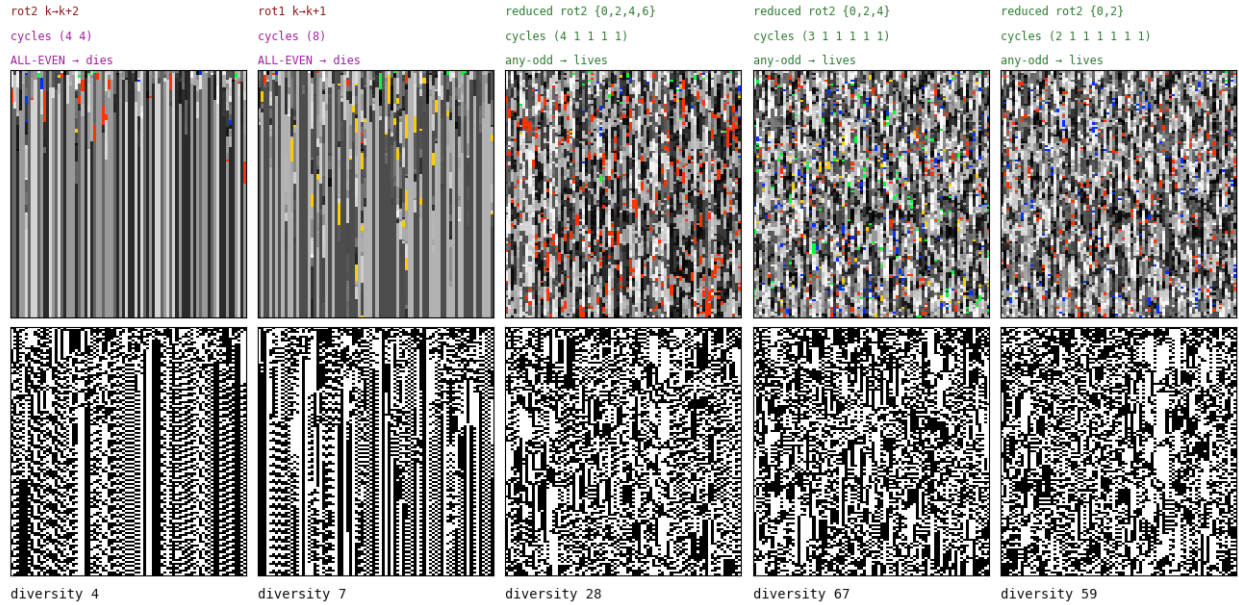


Figure 4

The parity dichotomy, by example (the full census of 20,256 orbits is summarised in the text). Five operators as genotype (top) and phenotype (bottom) over time. Left: the all-even operators rotate+2 and rotate+1 possess fixed points (Theorem 1) and, left to settle, the field collapses onto them—rotate+2 onto four rules, its $2^k = 2^2$ count. Right: the any-odd operators (reduced rotate+2, acting on a subset so the remaining positions are odd one-cycles) possess no fixed point, cannot settle, and stay diverse. This settling is the operator left to itself; a neighbour-blend interrupter reverses it for the gcd2 case (Figure 6, Fixed Points Are Not Attractors), and that reversal is the dynamical point.

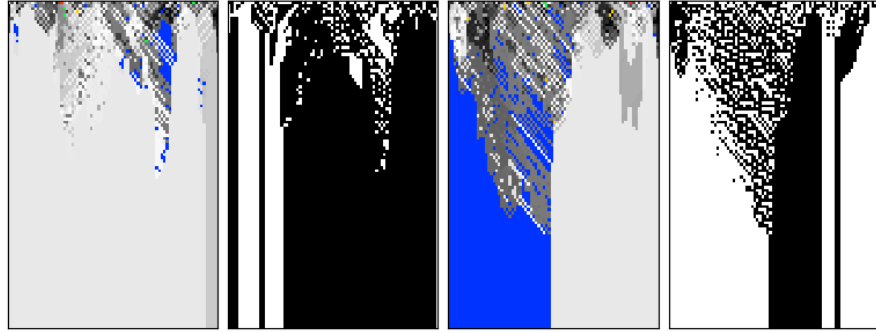
A family of operators, ranked by cycle structure (a · negative offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

offset -4

gcd 4 · four 2-cycles

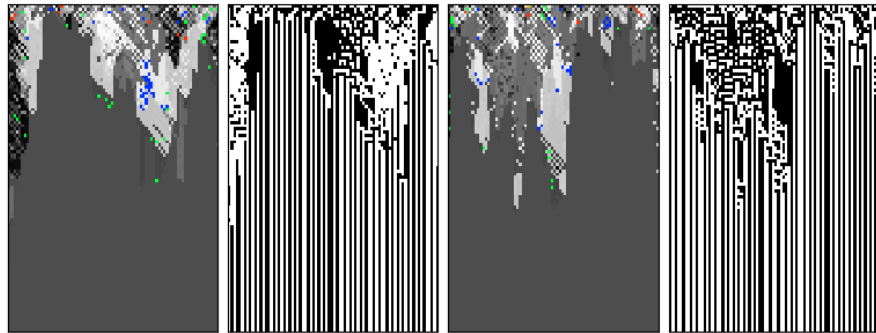
collapses ~t50



offset -3

gcd 1 · one 8-cycle

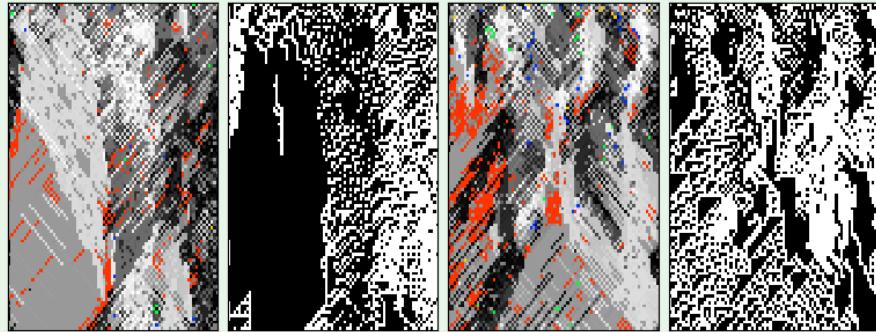
collapses ~t56



offset -2

gcd 2 · two 4-cycles

EDGE OF CHAOS (~t100)



offset -1

gcd 1 · one 8-cycle

collapses ~t46

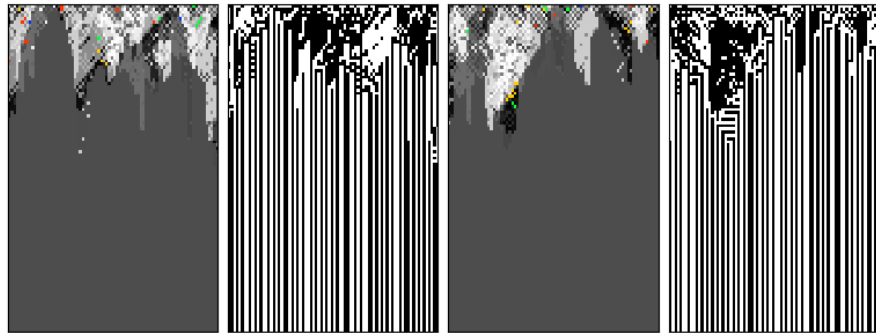


Figure 5

(a) *Negative offsets.* The rotation subfamily, ordered by offset. Each row is one operator $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$ with σ a wrap rotation, shown for two seeds as genotype (256-colour) and phenotype (black and white) over the first hundred generations. Every non-trivial rotation is all-even and so possesses fixed points (Theorem 1); yet only

A family of operators, ranked by cycle structure (b · positive offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

offset +1

gcd 1 · one 8-cycle

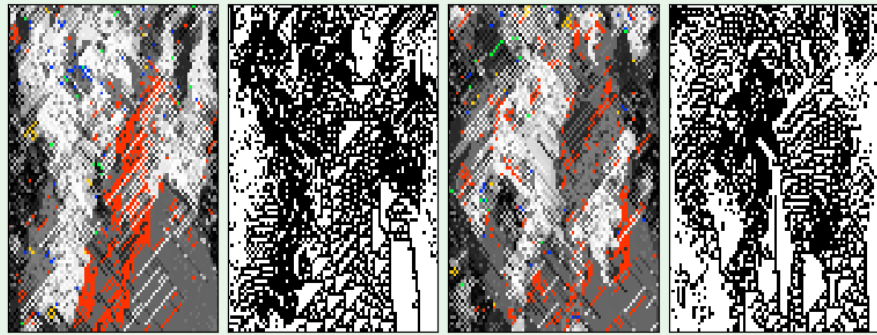
collapses ~t53



offset +2

gcd 2 · two 4-cycles

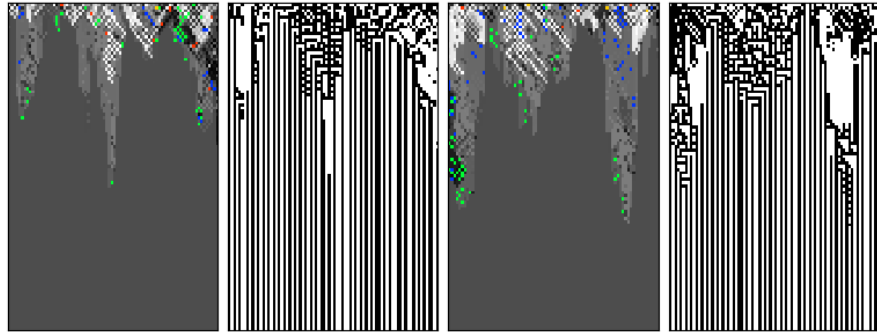
EDGE OF CHAOS (~t100)



offset +3

gcd 1 · one 8-cycle

collapses ~t48



offset +4

gcd 4 · four 2-cycles

collapses ~t50

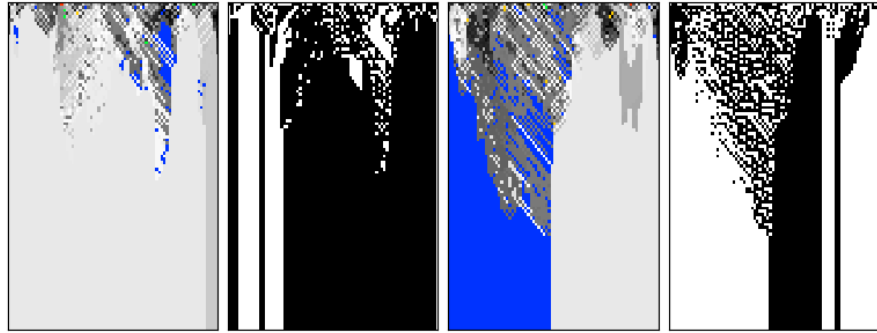
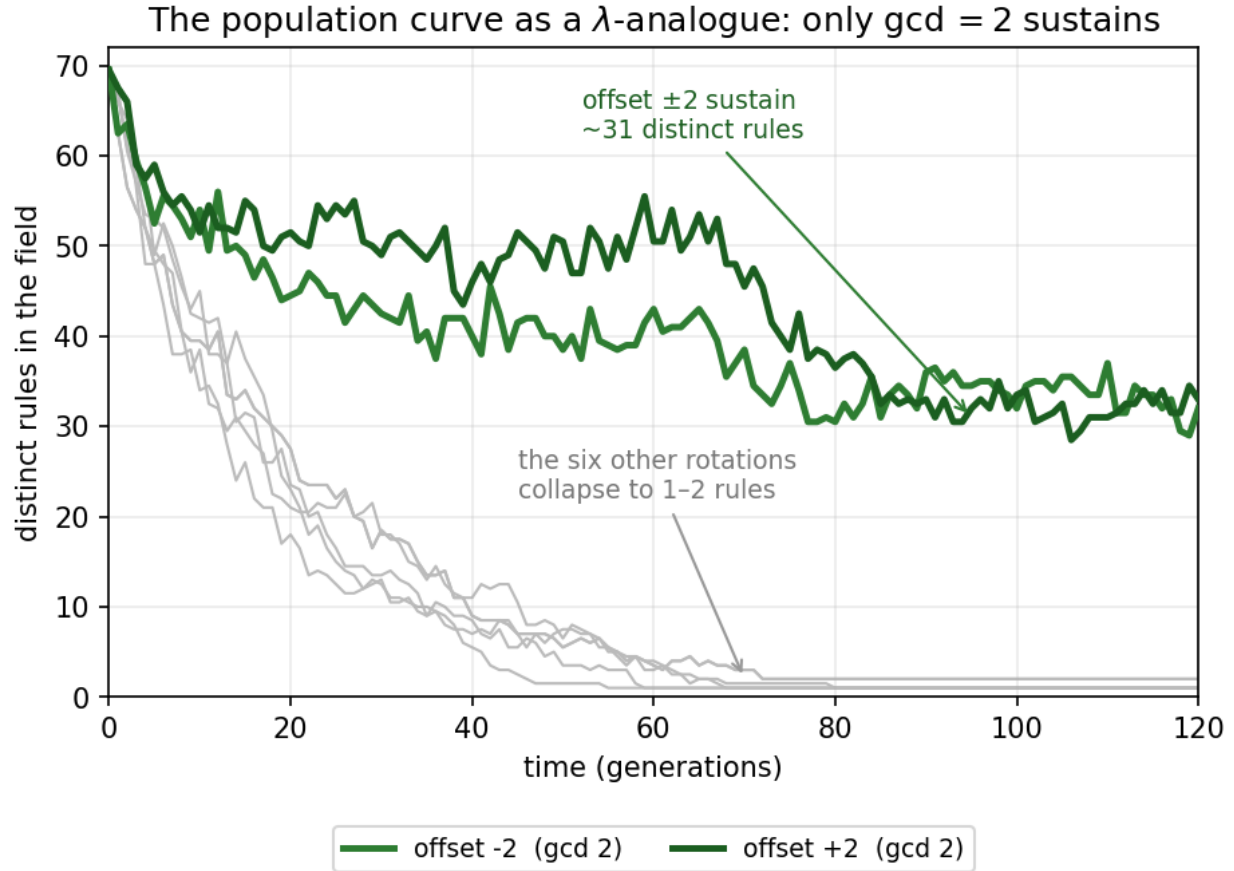


Figure 5

(b) *Positive offsets.* Continued from panel (a). The pattern is symmetric: offset +2 (gcd 2, two 4-cycles) sustains the high-diversity phase, the rest collapse. Possessing a fixed point and having the field stay near it are different properties, and this pair of pages is the whole paper in one figure.

**Figure 6**

A population observable: distinct rules in the field over time, for the eight wrap rotations (two seeds each). All begin at ≈ 70 ; only the two $\gcd 2$ operators, offset ± 2 (green), keep a large diverse population, while the six others collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure 5. The interrupting neighbour-blend is essential to the sustained case: without it these same operators, in the census runs, instead settle onto a handful of rules matching their 2^k count (rotate+2 onto four; Figure 4).

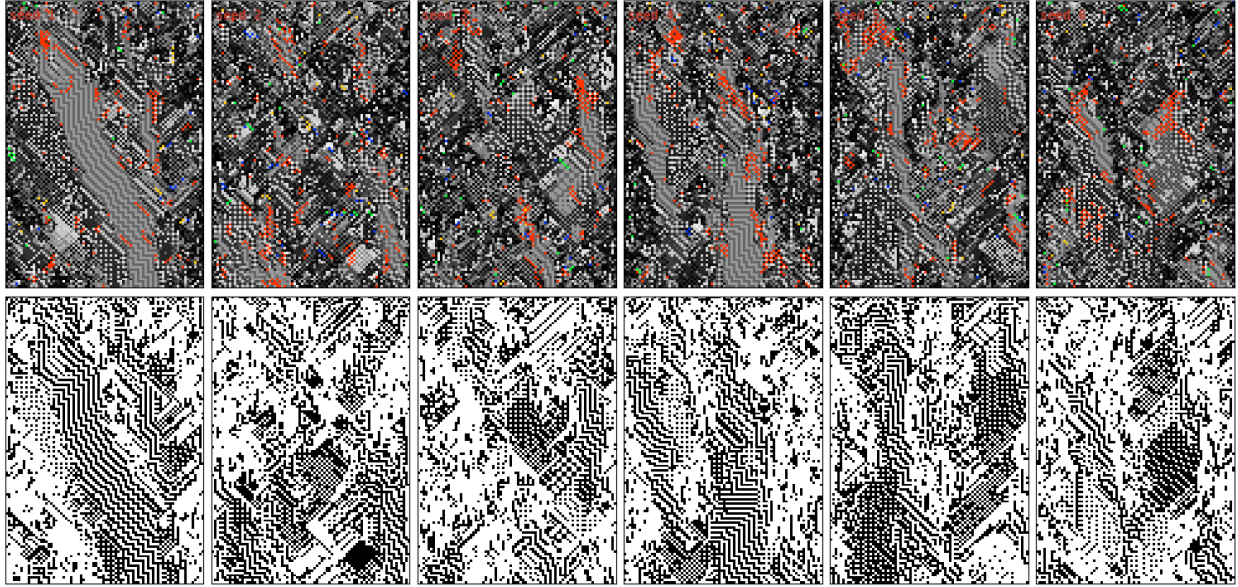
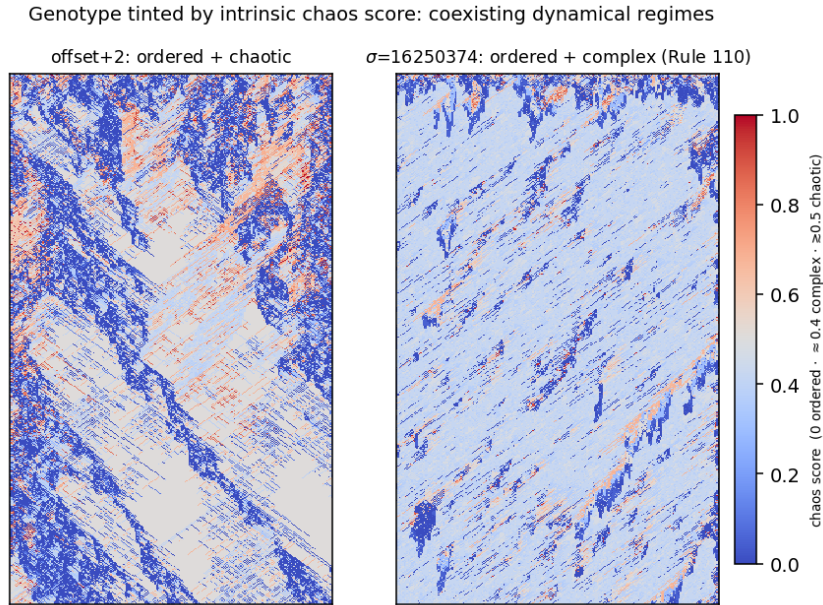


Figure 7

A river: a composed operator sustains a reproducible structure. The offset+2 propagator, composed with a phenotype-reading, four-candidate, first-match template construction (fallback: the all-zero rule), produces in each of six seeds a coherent band of alternating phenotype (bottom row) drifting through a disordered background (genotype top, six seeds; genotype coloured by rule as in Figures 1–5—greyscale by byte value, with well-known ECAs highlighted, red = Rule 110). The band’s trajectory varies with the seed; its presence does not. Where the bare rotations collapse or cycle (Figure 5), this composition holds an off-critical structure for the length of the run.

**Figure 8**

***The genotype as a map of dynamical regimes.** Genotype fields (fixed-boundary variant) tinted by each rule's intrinsic chaos score—its asymptotic activity as an isolated elementary CA (blue = ordered near 0, pale $\approx$ complex near 0.4, red = chaotic near 0.5 and above). The field resolves into coexisting regimes, and which active regime pairs with the ordered background depends on the operator: offset +2 mixes ordered and genuinely chaotic rules ($\approx 40\%$ chaotic at this seed), whereas $\sigma = 16250374$ is dominated by the complex Rule 110 ($\approx 48\%$; Figure 1) and pairs ordered with complex. The activity score orders the regimes but does not by itself separate complex from chaotic, a known-hard distinction.*

Order/chaos interface: fractal and size-stable for sustained operators, absent under collapse

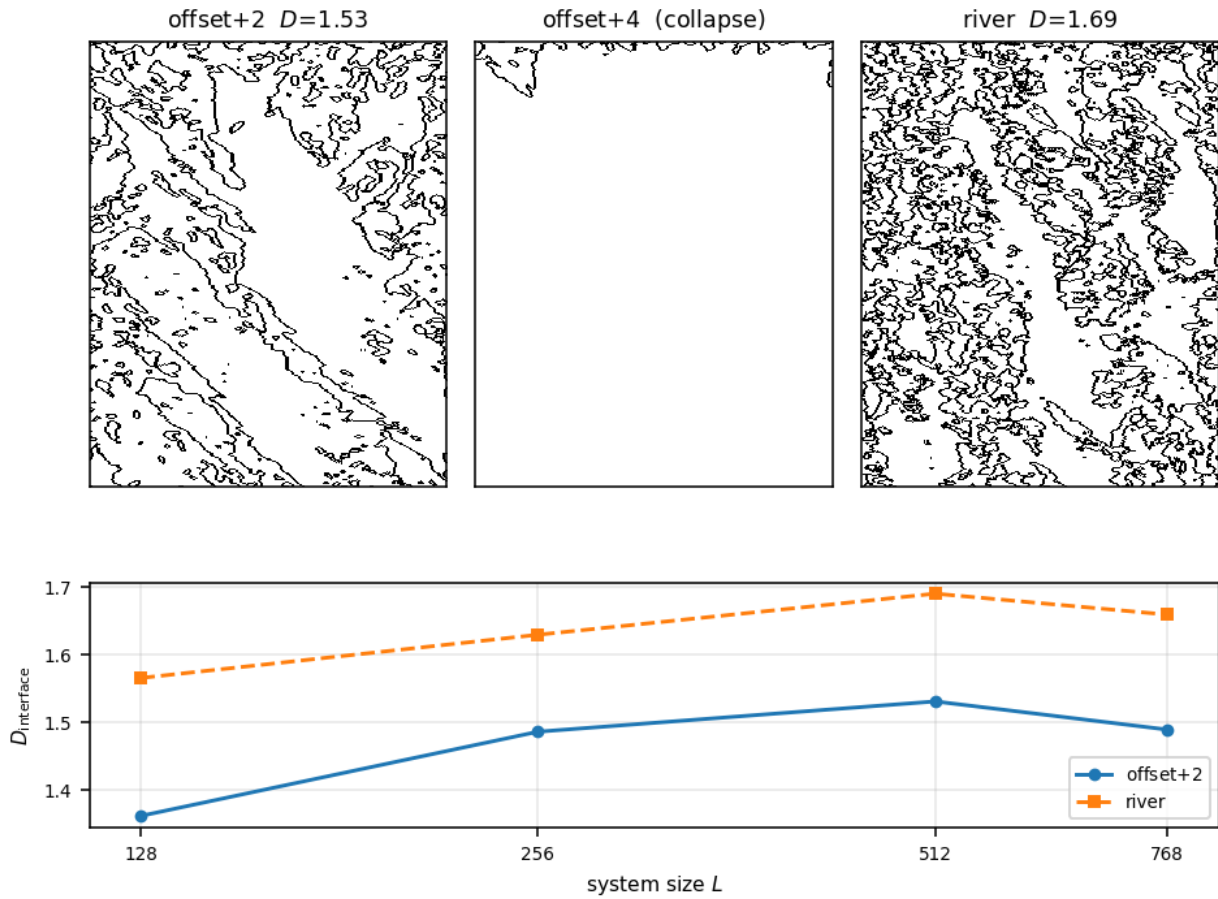


Figure 9

The order/chaos interface is fractal and size-stable for the sustaining operators. Top: the interface (black) extracted from the tinted field at $L = 256$. offset +2 carries a coherent fractal boundary ($D \approx 1.53$); offset +4 has collapsed (no interface); the phenotype-reading river carries a denser one ($D \approx 1.69$). Bottom: box-counting dimension against system size L ; the estimate stabilises rather than drifting. Exploratory: a handful of seeds, fixed-boundary variant.

References

- Aracena, J. (2008). Maximum number of fixed points in regulatory Boolean networks. *Bulletin of Mathematical Biology*, 70(5), 1398–1409.
- Corneli, J., & Maclean, E. (2015). The search for computational intelligence [v1].
- Crutchfield, J. P., & Young, K. (1989). Inferring statistical complexity. *Physical Review Letters*, 63(2), 105–108. <https://doi.org/10.1103/PhysRevLett.63.105>
- Dennunzio, A., Formenti, E., & Provillard, J. (2011). Non-uniform cellular automata: Classes, dynamics, and decidability [arXiv:1107.5228].
- Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3), 12–37. [https://doi.org/10.1016/0167-2789\(90\)90064-V](https://doi.org/10.1016/0167-2789(90)90064-V)
- Mitchell, M., Hraber, P. T., & Crutchfield, J. P. (1993). Revisiting the edge of chaos: Evolving cellular automata to perform computations. *Complex Systems*, 7(2), 89–130.
- Mori, T., Kudo, K., Namagawa, Y., Nakamura, R., Yamakawa, O., Suzuki, H., & Uesugi, T. (1998). Edge of chaos in rule-changing cellular automata. *Physica D: Nonlinear Phenomena*, 116, 275–282.
- OEIS Foundation Inc. (2011). Sequence A001818: Squares of double factorials; number of permutations of S_{2n} in which all cycles have even length [<https://oeis.org/A001818>; even-cycle interpretation dated 2011].
- Packard, N. H. (1988). Adaptation toward the edge of chaos. In J. A. S. Kelso, A. J. Mandell, & M. F. Shlesinger (Eds.), *Dynamic patterns in complex systems* (pp. 293–301). World Scientific.
- Pavlic, T. P., Adams, A. M., Davies, P. C. W., & Walker, S. I. (2014). Self-referencing cellular automata: A model of the evolution of information control in biological systems [arXiv:1405.4070].
- Riordan, J. (1968). *Combinatorial identities*. Wiley.

- Sakai, S., & Kanno, M. (2002). Structure of rule table and phase diagram of one dimensional cellular automata [arXiv:nlin/0204045].
- Schaller, M., & Svozil, K. (2025). Irreducible rules and equivalence classes of one-dimensional cellular automata [arXiv:2512.08117].
- Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4), 817–879. <https://doi.org/10.1023/A:1010388907793>
- Stanley, R. P. (1999). *Enumerative combinatorics, volume 2* [Problem 5.34(c)]. Cambridge University Press.
- Wuensche, A. (1999). Classifying cellular automata automatically: Finding gliders, filtering, and relating space-time patterns, attractor basins, and the Z parameter. *Complexity*, 4(3), 47–66.